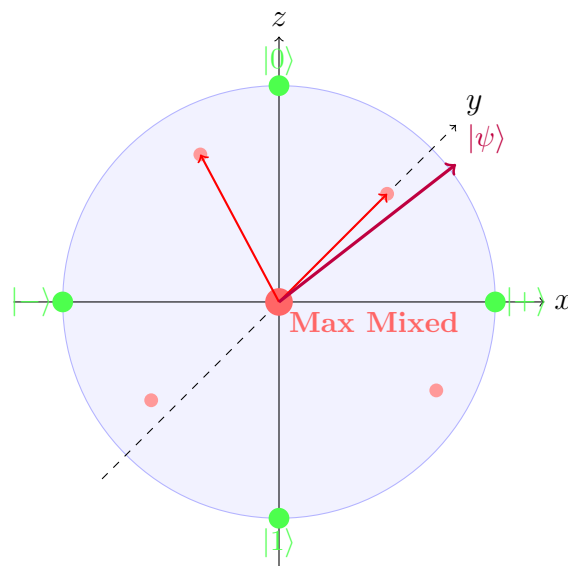


COMPREHENSIVE GUIDE TO DENSITY MATRICES

Pure States, Mixed States & Quantum Reality

COMPREHENSIVE LEARNING OBJECTIVES

1. **Conceptual Mastery:** Deep understanding of pure vs mixed quantum states
2. **Mathematical Fluency:** Expert density matrix calculations and proofs
3. **The Purity Test:** Master $\text{Tr}(\rho^2)$ as the ultimate quantum detector
4. **Bloch Sphere Visualization:** Geometric interpretation of quantum states
5. **Physical Applications:** Thermal states, decoherence, and quantum systems
6. **Problem-Solving Mastery:** Systematic approach to complex problems
7. **Advanced Topics:** Entanglement, partial traces, and quantum information



PURE STATES: $\text{Tr}(\rho^2) = 1$ (Sphere Surface)

MIXED STATES: $\text{Tr}(\rho^2) < 1$ (Sphere Interior)

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Course: Quantum Entanglement & Quantum Computing

A complete guide to understanding quantum states and density matrices

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1 Quantum States: The Foundation of Reality

1.1 The Quantum Information Problem

The Quantum Information Challenge

Classical physics assumes we can have complete information about any system. Quantum mechanics fundamentally challenges this assumption in several ways:

1. Heisenberg Uncertainty Principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Cannot know position and momentum simultaneously with perfect precision.

2. Quantum Measurement Disturbance Measuring one property destroys information about conjugate properties.

3. Environmental Decoherence Interaction with surroundings inevitably destroys quantum coherence.

4. Thermal Equilibrium At finite temperature, systems exist in statistical mixtures of energy states.

5. Entanglement Effects Measuring part of an entangled system leaves the remainder in a mixed state.

PHYSICS CONNECTION

Real-World Examples Where Mixed States Arise:

- **Quantum computers:** Decoherence limits computation time
- **Atoms at room temperature:** Thermal motion creates mixed states
- **Photons from stars:** Thermal radiation is inherently mixed
- **Biological systems:** Quantum effects in photosynthesis involve mixed states
- **Quantum cryptography:** Security depends on distinguishing pure from mixed states

1.2 Mathematical Framework

The Density Matrix Formalism

The density matrix ρ is a mathematical tool that describes quantum systems with incomplete information. It was introduced by Lev Landau in 1927 and has become fundamental to quantum mechanics.

Key Properties of Density Matrices:

1. **Hermitian:** $\rho^\dagger = \rho$ (ensures real eigenvalues)
2. **Positive semi-definite:** $\rho \geq 0$ (ensures non-negative probabilities)
3. **Unit trace:** $\text{Tr}(\rho) = 1$ (probability conservation)
4. **Physical interpretation:** $\rho_{ij} = \langle i | \rho | j \rangle$ represents quantum amplitudes

MATHEMATICAL PROOF

Why These Properties Are Necessary:

Hermiticity ($\rho^\dagger = \rho$): For any observable $\hat{A}$, the expectation value is:

$$\langle \hat{A} \rangle = \text{Tr}(\rho \hat{A})$$

This must be real for physical observables, which requires ρ to be Hermitian.

Positivity ($\rho \geq 0$): For any state $|\psi\rangle$, the probability is:

$$P = \langle \psi | \rho | \psi \rangle \geq 0$$

This requires all eigenvalues of ρ to be non-negative.

Unit Trace ($\text{Tr}(\rho) = 1$): The total probability must be:

$$\sum_i P_i = \sum_i \langle i | \rho | i \rangle = \text{Tr}(\rho) = 1$$

2 Pure States: Complete Quantum Information

2.1 Mathematical Definition and Properties

KEY CONCEPT

Pure State Definition

A quantum state is **pure** if it can be described by a single state vector $|\psi\rangle$.

Density Matrix Representation:

$$\rho = |\psi\rangle\langle\psi|$$

Fundamental Characteristic:

$$\text{Tr}(\rho^2) = 1$$

This is the definitive test for purity - it never fails!

MATHEMATICAL PROOF

Proof that $\text{Tr}(\rho^2) = 1$ for Pure States:

For any pure state $\rho = |\psi\rangle\langle\psi|$:

$$\rho^2 = (|\psi\rangle\langle\psi|)(|\psi\rangle\langle\psi|) \quad (1)$$

$$= |\psi\rangle\langle\psi|\psi\rangle\langle\psi| \quad (2)$$

$$= |\psi\rangle \cdot 1 \cdot \langle\psi| \quad (\text{since } \langle\psi|\psi\rangle = 1) \quad (3)$$

$$= |\psi\rangle\langle\psi| = \rho \quad (4)$$

Therefore: $\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1$

Alternative proof using spectral decomposition: For pure states, only one eigenvalue $\lambda_1 = 1$, all others are zero.

$$\text{Tr}(\rho^2) = \sum_i \lambda_i^2 = 1^2 + 0^2 + 0^2 + \dots = 1$$

2.2 Comprehensive Examples of Pure States

DETAILED EXAMPLE

Example 1: Computational Basis State $|0\rangle$

Step 1: State vector representation

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \langle 0| = (1 \ 0)$$

Step 2: Construct density matrix

$$\rho = |0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Step 3: Verify properties

- **Hermitian:** $\rho^\dagger = \rho$
- **Positive:** Eigenvalues are $\{1, 0\}$
- **Normalized:** $\text{Tr}(\rho) = 1 + 0 = 1$

Step 4: Purity test

$$\rho^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \rho$$

$$\text{Tr}(\rho^2) = 1$$

CONFIRMED PURE

Step 5: Physical interpretation This represents a qubit definitively in the "0" state with complete certainty.

DETAILED EXAMPLE

Example 2: Superposition State $|+\rangle$

Given: $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Step 1: Matrix representation

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \langle +| = \frac{1}{\sqrt{2}} (1 \ 1)$$

Step 2: Density matrix construction

$$\rho = |+\rangle\langle +| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} (1 \ 1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Step 3: Detailed purity verification

$$\rho^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \rho$$

$$\text{Tr}(\rho^2) = \text{Tr}(\rho) = \frac{1}{2}(1 + 1) = 1$$

CONFIRMED PURE

Step 4: Measurement analysis

- Probability of measuring $|0\rangle$: $P_0 = \langle 0|\rho|0\rangle = \frac{1}{2}$
- Probability of measuring $|1\rangle$: $P_1 = \langle 1|\rho|1\rangle = \frac{1}{2}$
- **Key insight:** Equal probabilities but definite phase relationship (coherent)

DETAILED EXAMPLE**Example 3: General Pure State**

Consider: $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$

This is the most general single-qubit pure state (Bloch sphere parameterization).

Step 1: Density matrix

$$\rho = \begin{pmatrix} \cos^2 \frac{\theta}{2} & \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{-i\phi} \\ \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{i\phi} & \sin^2 \frac{\theta}{2} \end{pmatrix}$$

Step 2: Eigenvalue analysis The eigenvalues are always $\{1, 0\}$ regardless of θ and ϕ .

Step 3: Purity verification

$$\text{Tr}(\rho^2) = 1$$

for all values of θ and ϕ .

Step 4: Bloch sphere representation

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

with $|\vec{r}| = 1$ always.

INTERACTIVE: Work This Out**Practice Problem 1: Verify Pure State Properties**

Given the state $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}|1\rangle$:

a) Verify this is a valid quantum state:

$$\langle\psi|\psi\rangle = \left| \frac{1}{\sqrt{3}} \right|^2 + \left| \frac{\sqrt{2}}{\sqrt{3}} \right|^2 =$$

b) Find the density matrix $\rho = |\psi\rangle\langle\psi|$:

$$\rho = \begin{pmatrix} \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \end{pmatrix}$$

c) Calculate $\text{Tr}(\rho^2)$ and verify purity:

$$\text{Tr}(\rho^2) =$$

$\underline{\hspace{1cm}} \Rightarrow \text{Pure/Mixed? } \underline{\hspace{1cm}}$

d) What are the measurement probabilities for $|0\rangle$ and $|1\rangle$?

$$P(|0\rangle) =$$

$\underline{\hspace{1cm}},$

$$P(|1\rangle) =$$

$\underline{\hspace{1cm}}$

2.3 Properties Unique to Pure States

Special Properties of Pure States

Pure states have several unique mathematical and physical properties:

1. Idempotent Property: $\rho^2 = \rho$

- This means $\rho^n = \rho$ for all positive integers n
- Physically: "Measuring a pure state repeatedly gives the same result"

2. Rank-1 Matrices: $\text{rank}(\rho) = 1$

- Only one non-zero eigenvalue (which equals 1)
- All information contained in a single vector

3. Maximum Information: Von Neumann entropy $S = 0$

- $S(\rho) = -\text{Tr}(\rho \log \rho) = 0$ for pure states
- Represents complete knowledge of the quantum system

4. Coherence Properties: Can exhibit quantum interference

- Off-diagonal elements can be non-zero and complex
- Phase relationships between amplitudes are preserved

ADVANCED TOPIC

Spectral Decomposition of Pure States

Any pure state density matrix has the spectral decomposition:

$$\rho = 1 \cdot |\psi\rangle\langle\psi| + 0 \cdot \sum_{i \perp \psi} |i\rangle\langle i|$$

where $\{|i\rangle\}$ form an orthonormal basis including $|\psi\rangle$.

Geometric Interpretation: Pure states correspond to rays in Hilbert space. The density matrix projects onto this ray with probability 1.

3 Mixed States: Quantum Reality

3.1 Physical Origins and Mathematical Definition

KEY CONCEPT

Mixed State Definition

A quantum state is **mixed** if it represents incomplete knowledge about a quantum system.

Mathematical Representation:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

where $p_i \geq 0$, $\sum_i p_i = 1$, and $\{|\psi_i\rangle\}$ are normalized states.

Fundamental Characteristic:

$$\text{Tr}(\rho^2) < 1$$

This inequality is the definitive signature of mixed states.

Four Primary Sources of Mixed States

1. Thermal Equilibrium

$$\rho_{\text{thermal}} = \frac{1}{Z} \sum_n e^{-E_n/k_B T} |n\rangle\langle n|$$

Systems at finite temperature exist in thermal mixtures.

2. Environmental Decoherence

$$\rho(t) = \mathcal{E}[\rho(0)]$$

Environment interactions destroy quantum coherence over time.

3. Partial Measurements If we measure only part of an entangled system, the remainder becomes mixed.

4. Incomplete State Preparation Classical uncertainty about which pure state was actually prepared.

3.2 Detailed Mixed State Examples

DETAILED EXAMPLE

Example 4: Classical Mixture of Basis States

Scenario: A quantum system is prepared with:

- Probability 0.7: state $|0\rangle$
- Probability 0.3: state $|1\rangle$

Step 1: Construct density matrix

$$\begin{aligned} \rho &= 0.7|0\rangle\langle 0| + 0.3|1\rangle\langle 1| \\ &= 0.7 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 0.3 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.3 \end{pmatrix} \end{aligned}$$

Step 2: Verify matrix properties

- **Hermitian:** $\rho^\dagger = \rho$ (real diagonal matrix)

- **Positive:** Eigenvalues $\{0.7, 0.3\} > 0$
- **Normalized:** $\text{Tr}(\rho) = 0.7 + 0.3 = 1$

Step 3: Purity calculation

$$\rho^2 = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.3 \end{pmatrix}^2 = \begin{pmatrix} 0.49 & 0 \\ 0 & 0.09 \end{pmatrix}$$

$$\text{Tr}(\rho^2) = 0.49 + 0.09 = 0.58 < 1$$

CONFIRMED MIXED**Step 4: Physical interpretation**

- This is a **classical mixture** - no quantum coherence
- Measurement outcomes are the same as the preparation probabilities
- No off-diagonal terms - no quantum interference possible

Step 5: Information content Von Neumann entropy: $S = -0.7 \log_2(0.7) - 0.3 \log_2(0.3) \approx 0.88$ bits

DETAILED EXAMPLE**Example 5: The Maximally Mixed State**

Definition: The state with minimum purity for a given dimension.

For a qubit: $\rho = \frac{1}{2}I = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Step 1: Purity analysis

$$\rho^2 = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{4}I$$

$$\text{Tr}(\rho^2) = \frac{1}{4}(1 + 1) = \frac{1}{2}$$

This is the **minimum possible purity** for any qubit state!

Step 2: Multiple decompositions The maximally mixed state can be decomposed in infinitely many ways:

Computational basis:

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

Hadamard basis:

$$\rho = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|$$

Any orthonormal basis:

$$\rho = \frac{1}{2}|\phi_1\rangle\langle \phi_1| + \frac{1}{2}|\phi_2\rangle\langle \phi_2|$$

Step 3: Physical properties

- **Maximum uncertainty:** All measurement outcomes are 50-50
- **Zero information:** Von Neumann entropy $S = 1$ bit (maximum for qubit)
- **Thermal interpretation:** Infinite temperature limit
- **Geometric:** Center of Bloch sphere ($\vec{r} = 0$)

DETAILED EXAMPLE**Example 6: Mixed State with Coherences**

Consider: $\rho = \begin{pmatrix} 0.6 & 0.2 + 0.1i \\ 0.2 - 0.1i & 0.4 \end{pmatrix}$

Step 1: Verify validity

- **Hermitian:** $\rho_{12} = (0.2 + 0.1i)^* = 0.2 - 0.1i = \rho_{21}$
- **Trace:** $\text{Tr}(\rho) = 0.6 + 0.4 = 1$
- **Positivity:** Need to check eigenvalues

Step 2: Eigenvalue calculation Characteristic polynomial: $\det(\rho - \lambda I) = \lambda^2 - \lambda + (0.6 \times 0.4 - |0.2 + 0.1i|^2)$

$$|0.2 + 0.1i|^2 = 0.04 + 0.01 = 0.05$$

$$\det(\rho - \lambda I) = \lambda^2 - \lambda + (0.24 - 0.05) = \lambda^2 - \lambda + 0.19$$

$$\lambda = \frac{1 \pm \sqrt{1 - 4(0.19)}}{2} = \frac{1 \pm \sqrt{0.24}}{2} \approx 0.755, 0.245$$

Both eigenvalues are positive

Step 3: Purity analysis

$$\text{Tr}(\rho^2) = \sum_i \lambda_i^2 = (0.755)^2 + (0.245)^2 \approx 0.630 < 1$$

MIXED

Step 4: Physical significance Unlike Example 4, this mixed state has:

- **Non-zero off-diagonal terms:** Some quantum coherence remains
- **Complex coherences:** Phase information is preserved
- **Partial decoherence:** Between pure and maximally mixed

INTERACTIVE: Work This Out**Practice Problem 2: Mixed State Analysis**

Consider the mixed state: $\rho = 0.4|0\rangle\langle 0| + 0.6|1\rangle\langle 1|$

a) Write the density matrix:

$$\rho = \begin{pmatrix} \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \end{pmatrix}$$

b) Calculate the purity:

$$\rho^2 = \begin{pmatrix} \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} & \underline{\hspace{1cm}} \end{pmatrix}$$

$$\text{Tr}(\rho^2) =$$

c) Compare with maximally mixed state: Maximally mixed purity = _____, This state's purity = _____

Which is "more mixed"? _____

d) Calculate Von Neumann entropy:

$$S = -0.4 \log_2(0.4) - 0.6 \log_2(0.6) =$$

_____ bits

3.3 Non-Uniqueness of Mixed State Decompositions

ADVANCED TOPIC

Fundamental Theorem: Non-Unique Decompositions

Unlike pure states, mixed states do not have unique decompositions into pure states. Any mixed state ρ can be written as:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

in infinitely many different ways.

Example: Three Different Decompositions of $\rho = \frac{1}{2}I$

Decomposition 1 (Z-basis):

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

Decomposition 2 (X-basis):

$$\rho = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|$$

Decomposition 3 (Y-basis):

$$\rho = \frac{1}{2}|i\rangle\langle i| + \frac{1}{2}|-i\rangle\langle -i|$$

where $|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ and $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$.

All three representations describe the same physical state!

COMMON MISTAKE

Common Misconception: Superposition vs Mixture

Students often confuse:

- **Quantum superposition:** $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ (PURE)
- **Classical mixture:** $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$ (MIXED)

Key Differences:

Property	Superposition	Mixture
Density Matrix	$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	$\frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Purity $\text{Tr}(\rho^2)$	1	$\frac{1}{2}$
Interference	YES	NO
Phase Information	YES	NO

Physical Distinction: Superposition has definite phase relationships; mixture has classical uncertainty.

4 The Purity Test: Ultimate Quantum Detector

4.1 Mathematical Foundation and Proof

KEY CONCEPT

The Universal Purity Test

For any density matrix ρ :

$$\text{Tr}(\rho^2) = \begin{cases} 1 & \text{if and only if } \rho \text{ represents a PURE state} \\ < 1 & \text{if and only if } \rho \text{ represents a MIXED state} \end{cases}$$

This test is:

- **Universal:** Works for any quantum system (qubits, atoms, molecules, etc.)
- **Definitive:** Never gives false results
- **Computable:** Can be calculated from any density matrix

MATHEMATICAL PROOF

Complete Proof of the Purity Test

Theorem: $\text{Tr}(\rho^2) = 1$ if and only if ρ is pure.

Proof ($\Rightarrow$): If ρ is pure, then $\text{Tr}(\rho^2) = 1$

If ρ is pure, then $\rho = |\psi\rangle\langle\psi|$ for some normalized $|\psi\rangle$.

$$\rho^2 = (|\psi\rangle\langle\psi|)(|\psi\rangle\langle\psi|) = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho$$

$$\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1$$

Proof ($\Leftarrow$): If $\text{Tr}(\rho^2) = 1$, then ρ is pure

Let ρ have spectral decomposition: $\rho = \sum_i \lambda_i |i\rangle\langle i|$ where $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$.

Then: $\text{Tr}(\rho^2) = \sum_i \lambda_i^2$

Since $\sum_i \lambda_i = 1$ and all $\lambda_i \geq 0$, by Cauchy-Schwarz inequality:

$$\left(\sum_i \lambda_i \right)^2 \geq \sum_i \lambda_i^2$$

$$1 \geq \sum_i \lambda_i^2 = \text{Tr}(\rho^2)$$

Equality $\text{Tr}(\rho^2) = 1$ holds if and only if only one $\lambda_j = 1$ and all other $\lambda_i = 0$.

This means $\rho = |j\rangle\langle j|$, which is pure.

Corollary: $\text{Tr}(\rho^2) < 1$ if and only if ρ is mixed.

4.2 The Purity Spectrum

Understanding the Purity Scale

The purity $\text{Tr}(\rho^2)$ provides a continuous measure of "mixedness":

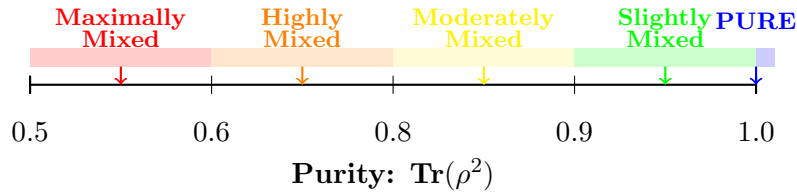
For Qubits:

- $\text{Tr}(\rho^2) = 1$: Pure state (maximum information)

- $\frac{1}{2} < \text{Tr}(\rho^2) < 1$: Partially mixed
- $\text{Tr}(\rho^2) = \frac{1}{2}$: Maximally mixed (minimum information)

For General d-dimensional Systems:

- $\text{Tr}(\rho^2) = 1$: Pure state
- $\frac{1}{d} < \text{Tr}(\rho^2) < 1$: Mixed state
- $\text{Tr}(\rho^2) = \frac{1}{d}$: Maximally mixed



4.3 Comprehensive Purity Problems

INTERACTIVE: Work This Out

Practice Problem Set: The Purity Detective

Use the purity test to classify each density matrix as pure or mixed:

Problem A: $\rho_A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Calculation: $\rho_A^2 =$ _____

$\text{Tr}(\rho_A^2) =$ _____ $\rightarrow$ Classification: _____

Problem B: $\rho_B = \begin{pmatrix} 0.8 & 0 \\ 0 & 0.2 \end{pmatrix}$

Calculation: $\rho_B^2 =$ _____

$\text{Tr}(\rho_B^2) =$ _____ $\rightarrow$ Classification: _____

Problem C: $\rho_C = \frac{1}{3} \begin{pmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}$

Calculation: $\rho_C^2 =$ _____

$\text{Tr}(\rho_C^2) =$ _____ $\rightarrow$ Classification: _____

Problem D: $\rho_D = \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$

First verify this is valid: $\text{Tr}(\rho_D) =$ _____

Eigenvalues: $\lambda_{1,2} =$ _____ (both positive? _____)

Calculation: $\rho_D^2 =$ _____

$\text{Tr}(\rho_D^2) =$ _____ $\rightarrow$ Classification: _____

DETAILED EXAMPLE

Example 7: Advanced Purity Calculation

Given: $\rho = \begin{pmatrix} 0.6 & 0.2e^{-i\pi/4} \\ 0.2e^{i\pi/4} & 0.4 \end{pmatrix}$

Step 1: Verify validity

- Hermitian: $\rho_{12} = 0.2e^{-i\pi/4} = (0.2e^{i\pi/4})^* = \rho_{21}^*$
- Trace: $\text{Tr}(\rho) = 0.6 + 0.4 = 1$

Step 2: Calculate ρ^2

$$\rho^2 = \begin{pmatrix} 0.6 & 0.2e^{-i\pi/4} \\ 0.2e^{i\pi/4} & 0.4 \end{pmatrix}^2$$

Computing each element:

$$(\rho^2)_{11} = 0.6 \times 0.6 + 0.2e^{-i\pi/4} \times 0.2e^{i\pi/4} = 0.36 + 0.04 = 0.40 \quad (5)$$

$$(\rho^2)_{12} = 0.6 \times 0.2e^{-i\pi/4} + 0.2e^{-i\pi/4} \times 0.4 = 0.2e^{-i\pi/4} \quad (6)$$

$$(\rho^2)_{21} = 0.2e^{i\pi/4} \times 0.6 + 0.4 \times 0.2e^{i\pi/4} = 0.2e^{i\pi/4} \quad (7)$$

$$(\rho^2)_{22} = 0.2e^{i\pi/4} \times 0.2e^{-i\pi/4} + 0.4 \times 0.4 = 0.04 + 0.16 = 0.20 \quad (8)$$

$$\rho^2 = \begin{pmatrix} 0.40 & 0.2e^{-i\pi/4} \\ 0.2e^{i\pi/4} & 0.20 \end{pmatrix}$$

Step 3: Calculate purity

$$\text{Tr}(\rho^2) = 0.40 + 0.20 = 0.60 < 1$$

MIXED STATE

Step 4: Physical interpretation This state has:

- Partial coherence (off-diagonal terms present)
- Definite phase relationship ($e^{i\pi/4}$ phase)
- Intermediate purity (between pure and maximally mixed)

5 Bloch Sphere: Geometric Quantum Mechanics

5.1 Complete Mathematical Framework

KEY CONCEPT

Universal Bloch Representation

Any qubit density matrix can be written as:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

where:

- $\vec{r} = (r_x, r_y, r_z)$ is the **Bloch vector**
- $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- $|\vec{r}| \leq 1$ always (fundamental constraint)

Geometric Classification:

- $|\vec{r}| = 1$: **Pure states** (on Bloch sphere surface)
- $|\vec{r}| < 1$: **Mixed states** (inside Bloch sphere)
- $|\vec{r}| = 0$: **Maximally mixed state** (sphere center)

How to Calculate Bloch Vector Components

Given a density matrix $\rho = \begin{pmatrix} a & b \\ b^* & c \end{pmatrix}$ with $a + c = 1$:

Method 1: Direct calculation

$$r_x = \text{Tr}(\rho\sigma_x) = 2\text{Re}(b) \quad (9)$$

$$r_y = \text{Tr}(\rho\sigma_y) = -2\text{Im}(b) \quad (10)$$

$$r_z = \text{Tr}(\rho\sigma_z) = a - c \quad (11)$$

Method 2: From matrix elements If $b = |b|e^{i\phi}$, then:

$$r_x = 2|b| \cos \phi \quad (12)$$

$$r_y = -2|b| \sin \phi \quad (13)$$

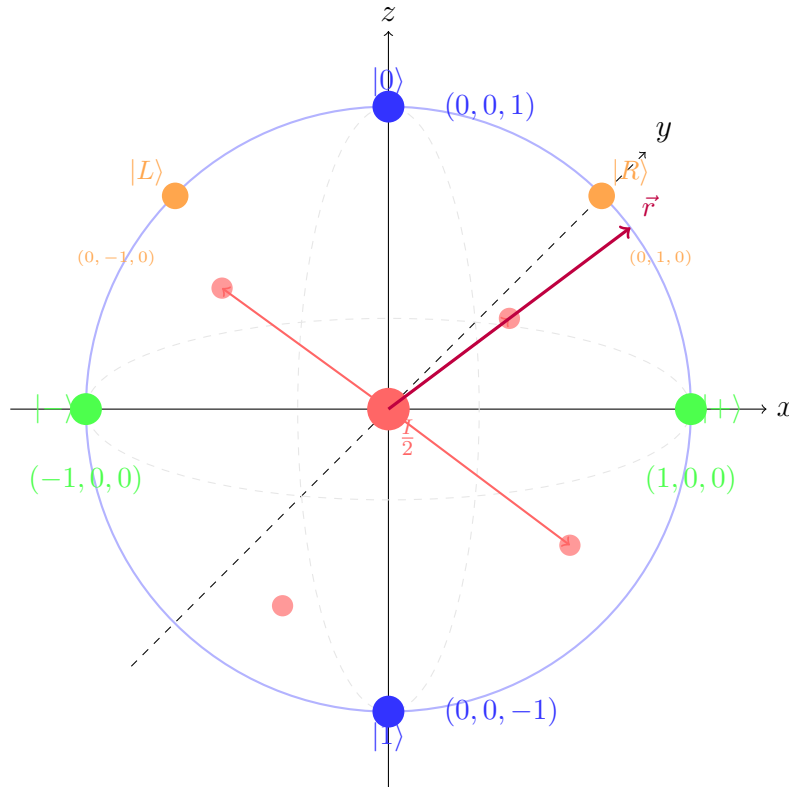
$$r_z = a - c \quad (14)$$

Verification: $|\vec{r}|^2 = r_x^2 + r_y^2 + r_z^2 = 4|b|^2 + (a - c)^2$

For pure states: $|\vec{r}| = 1$

For mixed states: $|\vec{r}| < 1$

5.2 Complete 3D Bloch Sphere Visualization



PURE STATES: Surface of sphere

MIXED STATES: Interior of sphere

$$\text{Purity} = \frac{1+|\vec{r}|^2}{2}, \quad \text{Tr}(\rho^2) = \frac{1+|\vec{r}|^2}{2}$$

5.3 Detailed Bloch Vector Examples

DETAILED EXAMPLE

Example 8: Complete Bloch Analysis

Given: $\rho = \begin{pmatrix} 0.7 & 0.3 - 0.1i \\ 0.3 + 0.1i & 0.3 \end{pmatrix}$

Step 1: Verify validity

- Hermitian: $(0.3 - 0.1i)^* = 0.3 + 0.1i$
- Trace: $0.7 + 0.3 = 1$
- Positivity check needed (eigenvalues)

Step 2: Calculate Bloch vector

$$r_x = 2\text{Re}(0.3 - 0.1i) = 2(0.3) = 0.6$$

$$r_y = -2\text{Im}(0.3 - 0.1i) = -2(-0.1) = 0.2$$

$$r_z = 0.7 - 0.3 = 0.4$$

Therefore: $\vec{r} = (0.6, 0.2, 0.4)$